

NETWORKED CONTROL

19/02/2020

Prof. Marcello Farina

SOLUTIONS

1) Consider the continuous-time system

$$\dot{x} = 2u$$

- A. Assuming that the system is controlled using a **continuous-time control law** of the type $u(t) = Kx(t)$, compute the range of values of K that make the closed-loop system asymptotically stable.
- B. Assuming that the system is controlled using a **digital control law** (i.e., with a sample-and-hold actuator) of the type $u_k = Kx_k$ and with sampling time $h = 0.2$ s, compute the range of values of K that make the closed-loop system asymptotically stable.
- C. Assume that the system is controlled using the **networked control system** sketched below.

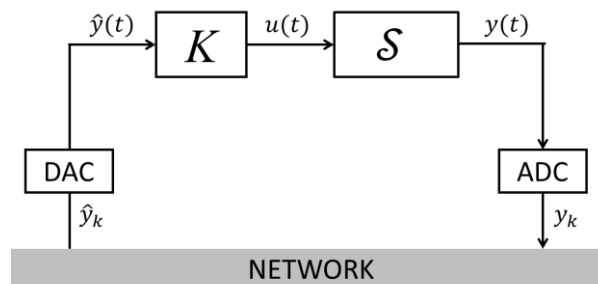


Figure 1: Networked control system

Assume that the sampling time is constant and equal to $h = 0.2$ s. Define the range of values of K that make the closed-loop system asymptotically stable in case the network delay is constant and equal to $\tau = 0.1$ s.

- D. Define succinctly the MATI (**maximal allowable transmission interval**) of a NCS.
- E. Assume again that the system is controlled using the networked control system sketched above, with constant sampling time and gain $K = -2.5$. Compute the MATI in the following two cases:
 - a. $\tau = 0$ s
 - b. $\tau = 0.1$ s

SOLUTIONS

- A. Assuming that the system is controlled using a continuous-time control law of the type $u(t) = Kx(t)$, the closed-loop system model is

$$\dot{x} = 2Kx$$

which is asymptotically stable if and only if $K < 0$.

- B. The corresponding discrete-time system is

$$x_{k+1} = x_k + 0.4u_k$$

Assuming that the system is controlled using a digital control law of the type $u_k = Kx_k$, the closed-loop system model is

$$x_{k+1} = (1 + 0.4K)x_k$$

which is asymptotically stable if and only if $-5 < K < 0$.

- C. Assuming that the system is controlled using a networked control system, with sampling time $h = 0.2$ s and delay $\tau = 0.1$ s, the NCS dynamics can be described by

$$z_{k+1} = \Phi(h, \tau)z_k$$

where $z_k = [x_k, \hat{y}_{k-1}]^T$ and

$$\Phi(h, \tau) = \begin{bmatrix} 1 - 2(h - \tau)K & 2K\tau \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 + 0.2K & 0.2K \\ 1 & 0 \end{bmatrix}$$

Using the Jury criterion, the matrix $\Phi(h, \tau)$ is Schur stable if and only if $-5 < K < 0$.

- D. See the lecture notes.

- E. If we set

- a. $K = -2.5$ and $\tau = 0$ s, then

$$\Phi(h, \tau) = \begin{bmatrix} 1 - 5h & 0 \\ 1 & 0 \end{bmatrix}$$

Using the Jury criterion, the matrix $\Phi(h, \tau)$ is Schur stable if and only if $0 < h < 0.4$ s. Therefore, the MATI is 0.4 s.

- b. $K = -2.5$ and $\tau = 0.1$ s, then

$$\Phi(h, \tau) = \begin{bmatrix} 1.5 - 5h & -0.5 \\ 1 & 0 \end{bmatrix}$$

Using the Jury criterion, the matrix $\Phi(h, \tau)$ is Schur stable if and only if $0 < h < 0.6$ s. Therefore, the MATI is 0.6 s.

2) What are the **benefits** of using **of control networks** over wired point-to-point connections for control purposes? What are the main **challenges** related to the use **of control networks**?

SEE THE LECTURE NOTES

3) What are the main differences between the **MAC protocols based on scheduling** and the **MAC protocols based on random multiple access**?

Describe the main **features and the properties** of one protocol for each of the two classes above.

SEE THE LECTURE NOTES